

Exercise 6: Library Management System

Linear Search checks each element one by one until the required element is found. It is simple to implement and works on both sorted and unsorted data.

Binary Search repeatedly divides the search space into two halves. It is faster than Linear Search but requires the data to be sorted beforehand.

2. Implementation:

```
class Book{
    int bookID;
    String title;
    String author;

    public Book(int bookID, String title, String author){
        this.bookID = bookID;
        this.title = title;
        this.author = author;
    }
}
```

Linear Search

```
public Book linearSearch(Book[] books, String title){
    for(Book book : books){
        if(book.title.equals(title)){
            return book;
        }
    }

    return null;
}
```

Binary Search

```
public Book binarySearch(Book[] books, String title){
    int low = 0;
    int high = books.length - 1;
    while(low <= high){
        int mid = low + (high - low) / 2;
        if(books[mid].title.equals(title)){
            return books[mid];
        }
    }
}
```

```
        if(books[mid].title.compareTo(title) < 0){
            low = mid + 1;
        }
        else{
            high = mid - 1;
        }
    }
    return null;
}
```

Analysis:

Linear Search:

- Best Case: $O(1)$
- Average Case: $O(n)$
- Worst Case: $O(n)$

Binary Search:

- Best Case: $O(1)$
- Average Case: $O(\log n)$
- Worst Case: $O(\log n)$

Binary Search is more efficient for large datasets because it reduces the search space after every comparison. However, it can only be used when the data is sorted. Linear Search is suitable for small datasets or unsorted collections of books.